Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. What is the principle by which the AEKF estimates $\Sigma_{\widetilde{w}}$?

1 / 1 point

- (selected) $\Sigma_{\widetilde{w}}$ is based on a history of innovations μ_k, where $\mu_k = z_k - h(\hat{x}_k^-, u_k, \bar{v}_k)$.
- $\Sigma_{\widetilde{w}}$ is based on a history of innovations μ_k, where $\mu_k = z_k - h(\hat{x}_k^+, u_k, \bar{v}_k)$.
- $\Sigma_{\widetilde{w}}$ is based on a history of residuals r_k, where $r_k = z_k - h(\hat{x}_k^-, u_k, \bar{v}_k)$.
- $\Sigma_{\widetilde{w}}$ is based on a history of residuals r_k, where $r_k = z_k - h(\hat{x}_k^+, u_k, \bar{v}_k)$.
- The AEKF does not estimate $\Sigma_{\widetilde{w}}$.

Correct
Yes, this is correct.

2. What is the principle by which the numerically robust AEKF estimates $\Sigma_{\widetilde{v}}$?

1 / 1 point

- $\Sigma_{\widetilde{v}}$ is based on a history of innovations μ_k, where $\mu_k = z_k - h(\hat{x}_k^-, u_k, \bar{v}_k)$.
- $\Sigma_{\widetilde{v}}$ is based on a history of innovations μ_k, where $\mu_k = z_k - h(\hat{x}_k^+, u_k, \bar{v}_k)$.
- $\Sigma_{\widetilde{v}}$ is based on a history of residuals r_k, where $r_k = z_k - h(\hat{x}_k^-, u_k, \bar{v}_k)$.
- (selected) $\Sigma_{\widetilde{v}}$ is based on a history of residuals r_k, where $r_k = z_k - h(\hat{x}_k^+, u_k, \bar{v}_k)$.
- The AEKF does not estimate $\Sigma_{\widetilde{v}}$.

Correct
Yes, this is correct.

3. What is the principle by which the AEKF estimates $\Sigma_{\tilde{x},0}$?

1 / 1 point

- $\Sigma_{\tilde{x},0}$ is based on a history of innovations μ_k, where $\mu_k = z_k - h(\hat{x}_k^-, u_k, \bar{v}_k)$.
- $\Sigma_{\tilde{x},0}$ is based on a history of innovations μ_k, where $\mu_k = z_k - h(\hat{x}_k^+, u_k, \bar{v}_k)$.
- $\Sigma_{\tilde{x},0}$ is based on a history of residuals r_k, where $r_k = z_k - h(\hat{x}_k^-, u_k, \bar{v}_k)$.
- $\Sigma_{\tilde{x},0}$ is based on a history of residuals r_k, where $r_k = z_k - h(\hat{x}_k^+, u_k, \bar{v}_k)$.
- (selected) The AEKF does not estimate $\Sigma_{\tilde{x},0}$.

Correct
Yes, this is correct.